

O-LEVELS Computer Science P2

Topic:

LINEAR SEARCH

DEFINITION: Scanning or searching in an array from start to End in order to find a target value

Target $\leftarrow$ 5

11	9	5	30	51
1	2	3	4	5

array values

index number

we will start comparing target with EACH VALUE in array one by one

Found

11
9
5
30
51

= Target x
= Target x
= Target ✓

CODE →

IF Target = array[i]

NO NEED TO SEARCH FURTHER IN THE ARRAY means you should BREAK the flow

x This is used when the data is sorted ✓

SORTED

5 | 9 | 11 | 30 | 51

UNSORTED

11 | 9 | 5 | 30 | 51

Time Complexity: $O(n)$ → Later discussion

DECLARE array: [1:5]: INTEGER

DECLARE Input, i : INTEGER

DECLARE Found: BOOLEAN

INPUT Input

FOR i ← 1 : 5

IF array[i] = Input

✓ Found ← true

END IF

IF FOUND = True

OUTPUT "value is true"

Question Initialize 2 arrays Username and Password with Random values. INPUT Username and pass.

Steps ① Find the username Entered by the user. If present get Index of the username entered by the user

② check if the password is present at the same index then OUTPUT "Logged in" else output verification failed.

Username

A	B	C	D	E	F
---	---	---	---	---	---

 4

Password

-	1	-	1	1	-
---	---	---	---	---	---

④ — save this index

① search username

————→ check if the password entered by the user matches the value of password array at index 4

1	2	3	④	5	1	2	3	④	5
Fahd	Amm	am	Mamba	Alib	123	321	111	666	222

- Input UserName of User ✓ $666 = 666$ ✓
Input Password of User
- save that index in a variable ✓
- does password entered by the user matches the value of $\text{password}[\text{index}]$ at same index

SOLUTION
ON THE
NEXT PAGE

```

DECLARE usernameList[1:5]: STRING —
DECLARE passwordList[1:5]: STRING —
DECLARE Username, UserPass: STRING —
DECLARE UserIndex: INTEGER
DECLARE Found: BOOLEAN

```

```

usernameList[1] ← "Aima"
usernameList[2] ← "Mamha"
usernameList[3] ← "Alichba"
usernameList[4] ← "Aimen"
usernameList[5] ← "Fahd"

```

```

passwordList[1] ← 111
passwordList[2] ← 222
passwordList[3] ← 000
passwordList[4] ← 999
passwordList[5] ← 666

```

```

OUTPUT "ENTER Username"

```

```

INPUT Username

```

```

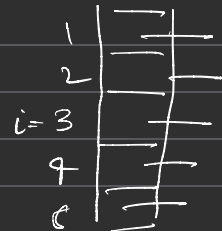
OUTPUT "ENTER Password"

```

```

INPUT Password

```



```

FOR i ← 1 to 5

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```

    IF Username = usernameList[i] THEN
        UserIndex ← i
        Found ← TRUE

```

```

    END IF

```

```

END FOR

```

IF Found = True

IF Password = PasswordList[UserIndex]
OUTPUT "Logged In"

ELSE OUTPUT "Wrong Password"

ELSE "User Name Not Found"

END IF

array name
PasswordList []
array ki index value